

Regression Analysis (I)

Final Exam.

June 19, 2017

1.(400) Consider a multiple linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \mathbf{I}\sigma^2)$. Let $\hat{\boldsymbol{\beta}}$ be the LSE of $\boldsymbol{\beta}$, $\mathbf{H}$ be the hat matrix, and $\mathbf{e}$ be the residual vector. Also, let $\hat{\boldsymbol{\beta}}^* = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$ be an unbiased estimator of $\boldsymbol{\beta}$ for some $p \times n$ matrix $\mathbf{D}$, where the rank of $\mathbf{D}$ is p .

(1)(50) Show that $Cov(\hat{\boldsymbol{\beta}}^*) - Cov(\hat{\boldsymbol{\beta}})$ is non-negative definite.

(2)(50) Express SSR in a quadratic form.

(3)(50) Find the distribution of SSR/σ^2 using the result of (2).

(4)(50) Show that $\hat{\mathbf{y}}$ and SSE are independent.

(5)(50) Express the LSE of $\boldsymbol{\beta}$ under $\mathbf{q}'\boldsymbol{\beta} = c$ in terms of $\hat{\boldsymbol{\beta}}$, $\mathbf{q}$, $\mathbf{X}$, and c . Here, $\mathbf{q}$ is p -vector.

(6)(50) Express $SSE(R) - SSE(F)$ in terms of $\hat{\boldsymbol{\beta}}$, $\mathbf{q}$, $\mathbf{X}$, and c , where $SSE(R)$ and $SSE(F)$ denote the error sum of squares under the reduced model in (5) and under the full model, respectively.

(7)(50) Let h_{ij} be the ij th component of $\mathbf{H}$, and e_j be the j th residual. Show that $\sum_{j=1}^n h_{ij}e_j = 0$, $\forall i$.

(8)(50) Using $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^t\mathbf{X})^{-1}\mathbf{x}_i e_i / (1 - h_{ii})$, show that the i th Cook's distance can be written as $C_i = r_i^2 h_{ii} / \{p(1 - h_{ii})\}$, where r_i is an internally Studentized residual.